



An efficient kinetic Monte Carlo scheme for computing Helmholtz free energy and entropy in bulk fluids and adsorption systems

Shiliang (Johnathan) Tan, D.D. Do*, D. Nicholson

School of Chemical Engineering, University of Queensland, St. Lucia, QLD 4072, Australia

ARTICLE INFO

Keywords:

Bulk fluids
Thermodynamics
Adsorption
Helmholtz free energy
Entropy
Monte Carlo simulation
Kinetic Monte Carlo simulation

ABSTRACT

We present an efficient kinetic Monte Carlo scheme to determine the Helmholtz free energy and entropy of bulk fluids and adsorption systems. The method is made possible because this technique enables the accurate determination of the chemical potential. The Helmholtz free energy, A , is obtained by integrating the chemical potential with respect to the number of molecules, at constant volume and temperature, and the entropy is then determined from the fundamental thermodynamic equation $A = E - TS$. The entropy of bulk argon is found to be in excellent agreement with values calculated from the established equation of state (*EOS*). In a system with two co-existing phases we show that our method can determine the surface tension at the Gibbs dividing surface, without recourse to the mechanical route of Irving, Kirkwood and Buff.

We show that the intrinsic integral molecular Helmholtz free energy and entropy of the adsorbed phase, corrected for the surface excess, are independent of the size of the simulation box for all chemical potentials tested.

The new procedure is illustrated for a range of gases commonly used in the characterization of porous solids (argon, nitrogen, carbon dioxide and ammonia) as adsorbates and a graphitic slit pore as the model adsorbent.

1. Introduction

The Metropolis Monte Carlo (*mMC*) simulation method, based on importance sampling, was ingeniously developed by scientists at the Los Alamos National Laboratory in the US, and has been widely used by scientists and engineers to determine equilibrium properties in numerous problems in chemistry and chemical engineering [1]. More recently, an alternative scheme based on the kinetic Monte Carlo (*kMC*) has been developed, and applied to a number of problems in colloid chemistry and to adsorption on surfaces and in confined spaces [2–8]. For example, various adsorbate transitions have been explored systematically where it was found that *kMC* offers distinct advantages over *mMC* [3] in systems with very high densities [9]. To ensure that equilibrium has been established, it is necessary to check the equality of chemical potential at all points in the system [9,10]. In traditional *mMC*, the chemical potential is usually determined by the Widom method [11,12], in which ghost particles are inserted into a number of frozen configurations. The accuracy of this method depends critically on the number of insertions, and decreases with increasing density, and is therefore very computationally intensive. By contrast, *kMC* provides an efficient route for calculating the chemical potential at all densities [10,13].

Other thermodynamic properties (*i.e.* Helmholtz free energy and entropy), are rarely studied in simulations, since there is no simple and satisfactory method for their determination in *mMC* [1]. The various methods that have been proposed [14–17], are either elaborate or cumbersome to apply [1]. Possibly the most beneficial feature of *kMC* is the accurate determination of the chemical potential [10] and we can take advantage of this to determine the Helmholtz free energy and the entropy. Moreover, in non-homogenous systems (*e.g.* phase co-existence and adsorption systems), the Helmholtz free energy and entropy can be decomposed into contributions from the constituent phases of the system, allowing an analysis of the specific contributions to free energy and the entropy from these phases.

Surface tension, associated with the boundary between two co-existing phases, is commonly determined by the mechanical route of Irving and Kirkwood [18] or Kirkwood and Buff [19]. Goujon et al. for example [20], employed this route to obtain the surface tension, and investigated the effects on the calculated energy of the cut-off radius, the long range correction, the three-body interaction, the box size and shifting the potential to zero at the cut-off radius.

In this work we use accurate values of chemical potential, determined by *kMC*, to find the Helmholtz free energy and entropy, and as an alternative to the mechanical method, to calculate the surface

* Corresponding author.

E-mail address: d.d.do@uq.edu.au (D.D. Do).

tension.

For adsorption systems, we have developed a means to determine the intrinsic Helmholtz free energy of the adsorbed phase. We show that this intrinsic molar excess free energy is lower than the molar free energy of a bulk liquid because of the stabilization of the adsorbed layer by the solid surface, but approaches the molar energy of the pure liquid as adsorbate coverage increases.

2. Theory

2.1. Kinetic Monte Carlo

A feature that distinguishes *kMC* from *mMC* is the absence of any acceptance criterion in a trial move. In *mMC*, many trial moves are rejected when the system energy is highly positive, due to repulsive overlap. In *kMC* all moves are accepted, even when there is a large overlap between molecules. Configurations with a strongly overlapping pair have an extremely short lifetime because one of them is highly likely to be selected in the move to generate the next configuration. The Rosenbluth scheme [1] governing this procedure is outlined briefly in the following sections.

2.1.1. Molecular energy and mobility rate

The molecular energy, u_i , of a molecule i is calculated as the sum of the pairwise intermolecular potential energies of molecule i with other molecules, $\varphi_{i,j}$, plus the total energy of interaction $\varphi_{i,S}$ with any solid atoms (constituting an external field if an adsorbent is present):

$$u_i = \sum_{\substack{j=1 \\ j \neq i}}^N \varphi_{i,j} + \varphi_{i,S} \quad (1)$$

The mobility v_i , of molecule i , which is a measure of the probability that it will move, is calculated from:

$$v_i = \exp(\beta u_i) \quad (2)$$

where $\beta = 1/k_B T$ and k_B is the Boltzmann constant. A molecule having a large molecular energy will therefore have a high mobility and possess a greater tendency to move. To select a candidate to move so as to maintain the stochastic nature of *kMC*, we apply a Rosenbluth scheme [1]. The time, during which the system stays in one configuration, is inversely proportional to the total mobility of the system, defined as the sum of all molecular mobilities (which is a measure of the *energetic state* of the current configuration):

$$R = \sum_{i=1}^N v_i = \sum_{i=1}^N \exp(\beta u_i) \quad (3)$$

We refer to this as the “total rate”. Eq. (3) states that the current configuration exists, on average, for a lifetime that is the inverse of the total rate, i.e. $\Delta t = 1/R$. Since many configurations will have the same total rate in the course of a simulation, the duration time follows a Poisson distribution law, i.e.

$$\Delta t = \frac{\ln(1/\xi)}{R} \quad (4)$$

where ξ is a random number ($0 < \xi < 1$).

2.2. Chemical potential

In previous studies of *kMC* [2,3] we have proved that the chemical potential, μ , can be calculated from the following equation:

$$\mu - \mu^{Intra} = k_B T \ln \left(\Lambda^3 \frac{\langle R \rangle}{V} \right) \quad (5)$$

where μ^{Intra} accounts for the rotation and vibration for polyatomic molecules, Λ is the thermal de Broglie wavelength, V is the volume of the simulation box, and $\langle R \rangle$ is the time average of the total rate calculated from

$$\langle R \rangle = \frac{\sum_{K=1}^M (\Delta t_K) R_K}{\tau}; \tau = \sum_{K=1}^M \Delta t_K \quad (6)$$

in which M is the number of configurations, τ is the total time of the simulation and $\Delta t_K/\tau$ is the fraction of the total time spent in configuration K (we use upper case subscripts to denote configuration number and lower case subscripts to denote a molecule). Substituting Eq. (4) into Eq. (6), we get:

$$\langle R \rangle = \frac{\sum_{K=1}^M \frac{\ln(1/\xi_K)}{R_K} R_K}{\tau} = \frac{M}{\tau} = \frac{1}{\langle \tau \rangle} \quad (7)$$

in which we have used $\sum_{K=1}^M \ln(1/\xi_K) = M$. Here $\langle \tau \rangle$ is the average lifetime of a configuration over the whole simulation, and is a function of the number of molecules in the system at a given volume and temperature [10]. To understand the average lifetime, we consider the extreme case of one molecule in the system; in this case, the total rate is 1, therefore the total time is M and hence the average lifetime is 1. When there are more molecules the average lifetime of a configuration for a stable system is less than 1. The total rate R is a measure of the interactions among the various entities in the system, and therefore it is informative to decompose the chemical potential into two contributions: one is the ideal gas contribution and the other is the excess contribution, which accounts for the energy of interaction between the entities (molecules and surface atoms). We rewrite Eq. (5) as the sum of the ideal gas and excess chemical potentials as follows:

$$\mu - \mu^{Intra} = \mu^{IG} + \mu^{ex} = k_B T \ln \left(\Lambda^3 \frac{N}{V} \right) + k_B T \ln \left(\frac{\langle R \rangle}{N} \right) \quad (8)$$

$\langle R \rangle/N$ is the average molecular mobility over the whole course of a simulation. Using Eq. (7), we express the excess chemical potential in terms of the average lifetime:

$$\beta \mu^{ex} = -\ln(N \langle \tau \rangle) \quad (9)$$

Since the mean residence time, $\langle \tau \rangle$, is a function of N for a given volume and temperature, the RHS of the above equation is a function of N .

2.3. Helmholtz free energy

In a closed system, equilibrium is said to have been reached when the Helmholtz free energy has reached its minimum (see Appendix A for a derivation). For an isotropic homogenous system, the Helmholtz free energy is calculated from the following fundamental thermodynamic equation, relating the Helmholtz free energy to the Gibbs free energy and the work, PV :

$$A = G - PV \quad (10)$$

where the Gibbs free energy can be computed from the chemical potential, $G = \mu N$, and the pressure P is calculated via the virial route. However, in a nonhomogeneous system, the calculation of the pressure tensor is extremely cumbersome, and therefore, we determine the

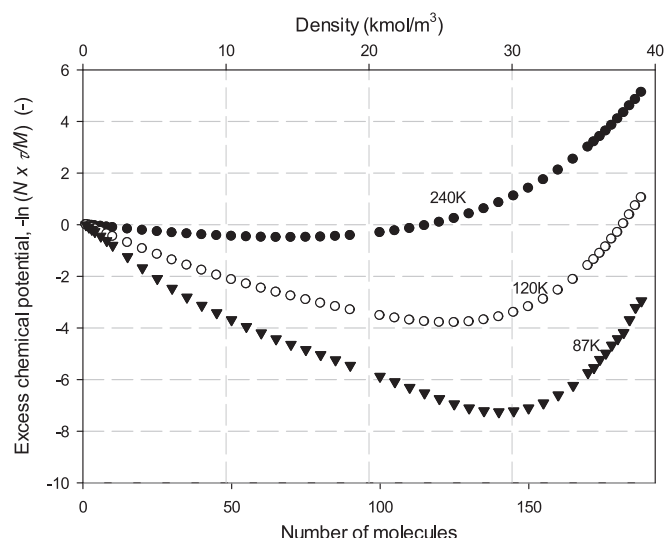


Fig. 1. Plot of the excess chemical potential scaled by kT , $\beta\mu^{\text{ex}} = -\ln(N\langle\tau\rangle)$ against the number of molecules and density for bulk argon at 87 K, 120 K and 240 K in a 2 nm cubic box.

Helmholtz free energy via a thermodynamic route without the recourse to the pressure tensor. The thermodynamic route uses the fact that the chemical potential is a partial derivative of the Helmholtz free energy with respect to the number of molecules, at constant volume and temperature:

$$\beta A = \int_1^N \mu(N) dN \quad (11)$$

In the same way as for the chemical potential, we decompose the Helmholtz free energy into ideal gas and excess contributions:

$$A = A^{\text{IG}} + A^{\text{ex}} \quad (12)$$

where these are given by:

$$\beta A^{\text{IG}} = \int_1^N \mu^{\text{IG}} dN; \beta A^{\text{ex}} = \int_1^N \mu^{\text{ex}} dN = - \int_1^N \ln(N\bar{\tau}) dN \quad (13)$$

in which we have used Eq. (9). The ideal gas Helmholtz free energy can be obtained by analytically integrating the ideal gas chemical potential to get $A^{\text{IG}} = kT [\ln(\Lambda^3 N/V) - 1]$, and the excess Helmholtz free energy can be obtained by numerical quadrature, exploiting the accuracy of the chemical potential obtained from kMC. We show in Fig. 1 a plot of the excess chemical potential, $[-\ln(N\langle\tau\rangle)]$, as a function of N for bulk argon at three temperatures, covering the sub- and supercritical regions. The smooth behaviour of this function allows us to evaluate the integral in Eq. (13) reliably. Given the Helmholtz free energy, the integral molecular Helmholtz free energy is:

$$\bar{A} = A/N \quad (14)$$

2.4. Entropy

The fundamental thermodynamic equation for the internal energy, $dE = TdS - pdV + \mu dN$, reduces to $dE - TdS = 0$ for a closed canonical ensemble at constant V and N . This leads to the definition of the Helmholtz free energy in terms of the internal energy and the entropy, $A = E - ST$. The internal energy is the sum of the kinetic and configurational energies, i.e. $E = E_K + U$, from which we can obtain the entropy:

$$S = \frac{E_K + U - A}{T} \quad (15)$$

In Eq. (15) the kinetic energy for a monoatomic species is

$E_K = (3/2)N k_B T$, and the configurational energy and Helmholtz free energy are calculated from the simulation. For polyatomic species, we present $S - E^{\text{Intra}}/T$. The integral molecular entropy is then $\bar{S} = S/N$.

2.5. Surface tension

The Helmholtz free energy can also be used to derive the surface tension when two phases coexist in the system, for example at vapour-liquid equilibrium. When the interface between the two phases is planar, the Gibbs construction locates the interface as the boundary between two regions of uniform gaseous and liquid density such that the number of molecules in the system is equal to that in the actual system [21]. The Helmholtz free energy of the system is then the sum of three terms: (1) the Helmholtz free energy of the uniform phase I, containing N^I molecules, (2) the Helmholtz free energy of the uniform phase II, containing $N^{\text{II}} = N - N^I$ molecules, and (3) the excess Helmholtz free energy associated with the interface. The latter must be positive to sustain the interface. Thus, we can write:

$$A = A^I + A^{\text{II}} + A^{\text{ex}} \quad (16)$$

where the Helmholtz free energies of the two uniform phases are given by

$$A^I = N^I (\bar{A})^I; A^{\text{II}} = N^{\text{II}} (\bar{A})^{\text{II}} \quad (17)$$

where $(\bar{A})^I$ and $(\bar{A})^{\text{II}}$ are the molecular Helmholtz free energies of the homogeneous phases I and II, respectively, determined from simulations of the bulk fluids of these two phases. Therefore, the excess Helmholtz free energy can be calculated from Eq. (16), and the surface tension of the interface, γ , can be calculated from

$$A^{\text{ex}} = \gamma \Sigma^{\text{I,II}} \quad (18)$$

where $\Sigma^{\text{I,II}}$ is the area of the interface separating the two phases.

2.6. Adsorption system

In an adsorption system, we can determine both the Helmholtz free energy of the whole system, and the intrinsic Helmholtz free energy of the adsorbed phase. The number of molecules associated with the adsorbed phase is normally calculated as an excess with respect to a reference amount, which is taken to be the number of molecules occupying a reference volume of the same density as that of the bulk phase. Here we adopt the same definition to maintain consistency between the

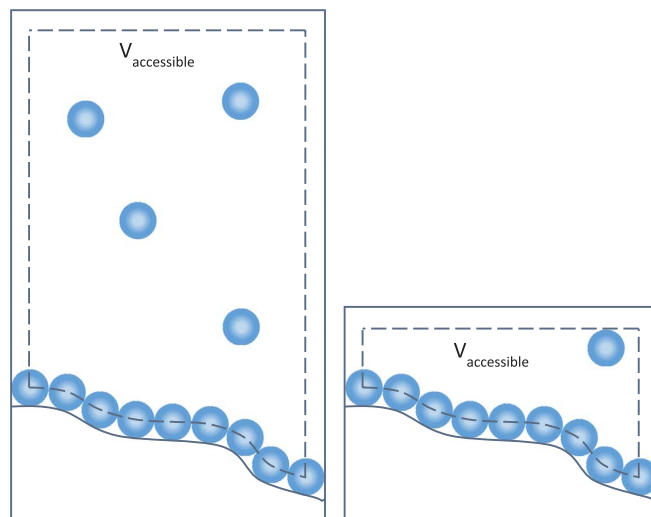


Fig. 2. A schematic diagram of adsorption showing two different sizes but having the same surface area.

Table 1
Molecular parameters.

Fluid	Parameter	Symbols	Unit	Value
Argon [23]	Argon	σ	nm	0.3405
		ϵ/k	K	119.8
Nitrogen, TraPPE [24]	N	σ	nm	0.331
		ϵ/k	K	36
		q	E	-0.482
	Additional charge, M	q	E	0.964
	Atomic distance	D_{M-N}	nm	0.055
Carbon Dioxide, TraPPE [24]	O	σ	nm	0.305
		ϵ/k	K	79
		q	E	-0.35
	C	σ	nm	0.28
		ϵ/k	K	27
		q	E	0.7
	Atomic distance	D_{C-O}	nm	0.116
Ammonia [25]	N	σ	nm	0.3385
		ϵ/k	K	170
		q	E	-1.035
	H	q	E	0.345
	Atomic distance	D_{N-H}	nm	1.00685, 1.0124, 1.0124
	Angle	Angle H-N-H	°	106

excess amount and the excess Helmholtz free energy.

We begin with the excess amount defined as:

$$N = N^{ex} + N^{Ref} \quad (19)$$

where the reference number of molecules is $N^{Ref} = \rho_B V^{ref}$, where ρ_B is the density of the bulk gas phase having the same chemical potential as the adsorption system, and V^{ref} is the volume accessible to the adsorbate [22]. With this choice of the reference amount, the excess amount, N^{ex} , is constant, irrespective of how large the gas phase is as illustrated schematically in Fig. 2.

In the spirit used to calculate the excess amount in a bulk fluid, we define the excess for the Helmholtz free energy (A), the entropy (S) and the configurational energy (U) by:

$$X = X^{ex} + X^{Ref} \quad (20)$$

where X is A , S or U . The reference quantity X^{Ref} is that of the bulk gas phase occupying the reference volume at the same density as the bulk surroundings; for example the reference configurational energy is:

$$U^{Ref} = N^{Ref} (\bar{U})^G \quad (21)$$

Here $(\bar{U})^G$ is the molecular configurational energy of the bulk gas having the same chemical potential as that of the system.

Knowing the excess quantity, the integral molecular quantity of the adsorbed phase is:

$$\bar{X}^{ex} = X^{ex} / N^{ex} \quad (22)$$

2.7. Molecular parameters

Argon, nitrogen, carbon dioxide and ammonia, are used to illustrate

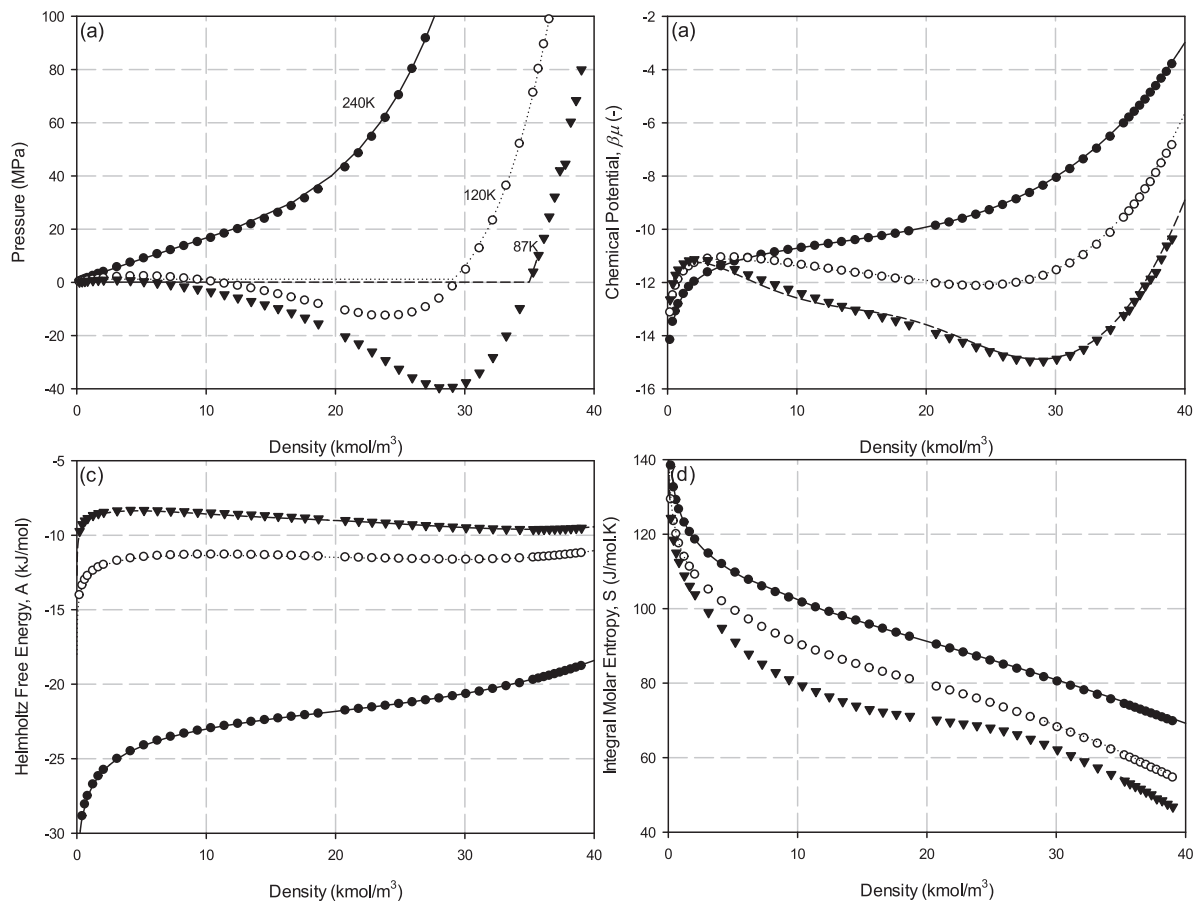


Fig. 3. (a) Pressure, (b) chemical potential, (c) integral molar Helmholtz free energy and (d) integral molar entropy of argon at 87 K, 120 K and 240 K in a 2 nm cubic box. The lines represent Equation of State (EOS) of Johnson et al. [30] and the symbols represent the simulated data.

the method. The molecular parameters are listed in Table 1. To simulate the bulk phase, we used a 2 nm cubic simulation box (unless stated), and periodic boundary conditions are imposed at all boundaries. The cut-off radius was taken as half the shortest length of the box side.

To study the thermodynamics of phase transition on a surface, we used an example of nitrogen adsorption at 77 K on a corrugated graphite surface, modelled with the Fourier approximation equation developed by Steele [26]. In addition to this corrugation, we took into account the differences in the polarizabilities of the carbon atom, parallel and vertical to the surface, as suggested by Carlos and Cole [27], details of this graphite molecular model can be found in Prasetyo et al. [28]. As a second example we took adsorption of argon at 87 K in a graphitic slit pore, where we show the thermodynamics of capillary condensation. This pore is connected to the bulk gas surrounding at both ends. The pore wall consists of three graphene layers and the solid-fluid potential energy is calculated with the Bojan-Steele equation, with the molecular parameters for a carbon atom in a graphene layer, $\sigma_{CC} = 0.34\text{ nm}$, $\epsilon_{CC}/k_B = 28\text{ K}$ and the interlayer spacing Δ of 0.3354 nm [29]. The cross-parameters were calculated from the Lorentz-Berthelot mixing rule.

3. Results

3.1. Bulk fluid

We show in Fig. 3 the four basic thermodynamic variables: pressure, chemical potential, molar Helmholtz free energy (via the thermodynamic route) and molar entropy of the bulk fluid argon at sub-critical temperatures of 87 K, 120 K and a supercritical temperature of 240 K. The simulated results from our method (with long range correction added to the molecular energy) are in excellent agreement with the EOS of Johnson et al. [30] and with the results obtained earlier by Fan et al. [10]. The percentage error of the chemical potential obtained with *kMC* scheme is typically less than 0.1%, showing the accuracy of the chemical potential. The results for other fluids are presented in Appendix B to demonstrate that the method is applicable equally to polyatomic and polar fluids.

In Fig. 4, we compare the Helmholtz free energy determined by the

virial route with the thermodynamic route, and note that there is excellent agreement. However, the success of the virial route depends critically on the determination of the pressure, which may not be accurate for dense systems. This is illustrated in Fig. 4a where the Helmholtz free energy of the dense phase at 87 K is fairly noisy, but if the pressure is determined with a sufficiently long simulation, it is able to produce a more accurate result than the thermodynamic route, since this avoids the accumulation of errors in the process of integration (Fig. 4b). In addition, the virial route is strictly valid only for uniform systems and the thermodynamic route is the only method available for non-uniform system. An accurate determination of bulk fluid properties is important in our assessment of the thermodynamic properties of the adsorption systems as we demonstrate below.

3.1.1. Effect of simulation box size

The pressure and the chemical potential shown in Fig. 3 exhibit a typical van der Waals (*vdW*) loop for temperatures (87 K and 120 K) below the bulk critical temperature. In Fig. 5 we explore the effects of the box size on the *vdW* behaviour for the four thermodynamic variables: pressure, chemical potential, molar Helmholtz free energy and molar entropy. The box size clearly affects the evolution of the *vdW* loop, notably for the chemical potential: for small box sizes, the *vdW* loop has a clear S-shape because of the small fluctuations and because the number of molecules in the system is not sufficient to form stable co-existing phases. The regions that show an increase in the chemical potential with respect to the number of molecules are either stable or metastable, while the region that shows a decrease is unstable. This S-shape is similar to that observed in the classical work of Hansen and Verlet [31] who suppressed fluctuations by dividing the box into regions and constraining the variation in the number of molecules in those regions. As the box size is increased, the *vdW* loop changes from a sigmoid shape to a step because the system forms two stable phases as evidenced by the constant chemical potential within the *vdW* loop. The evolution of the system into two stable co-existing phases has been studied by a number of workers [32,33]. Interestingly, the molar entropy, is not affected by the system size in the stable regions of gas and liquid phases. The entropy decreases with the number of molecules as expected because of the reduction in the degree of freedom as the

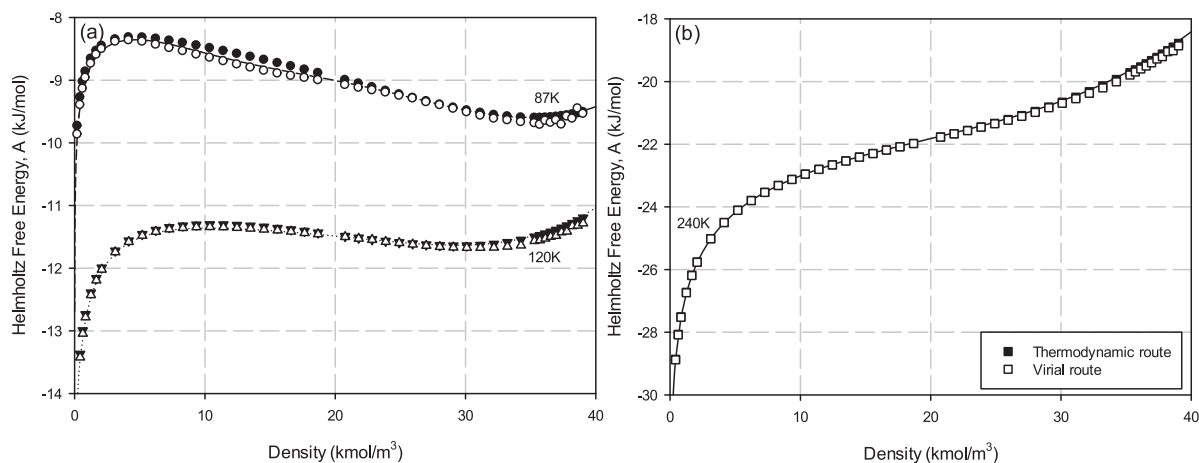


Fig. 4. The Helmholtz free energies determined via the virial (unfilled) and thermodynamic (filled) routes for argon at (a) 87 K (circle), 120 K (triangle) and (b) 240 K (square). The lines represent Equation of State (EOS) of Johnson et al. [30].

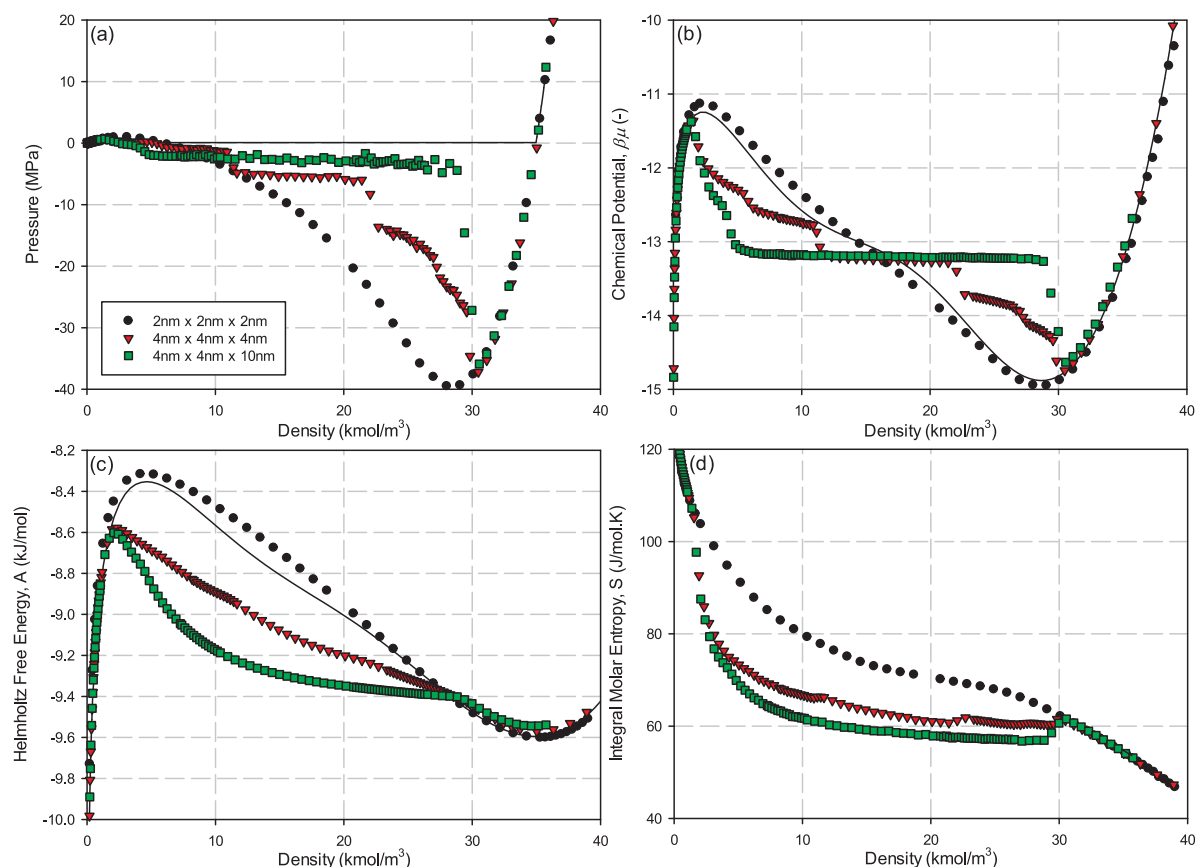


Fig. 5. (a) Pressure, (b) chemical potential, (c) Helmholtz free energy and (d) integral and differential molar entropy of argon at 87 K. The lines represent the EOS of Johnson et al. [30] and symbols represents the simulated data of different size systems from smallest (circle) to largest (square).

Table 2

Surface tension of argon.

Values are in mN/m	Surface tension	Standard deviation
This work at 87 K	15.70	0.30
Simulation at 85 K, Goujon et al. [20]	15.37	0.33
Experiment at 87 K, Lemmon and Penoncello [34]	12.607	

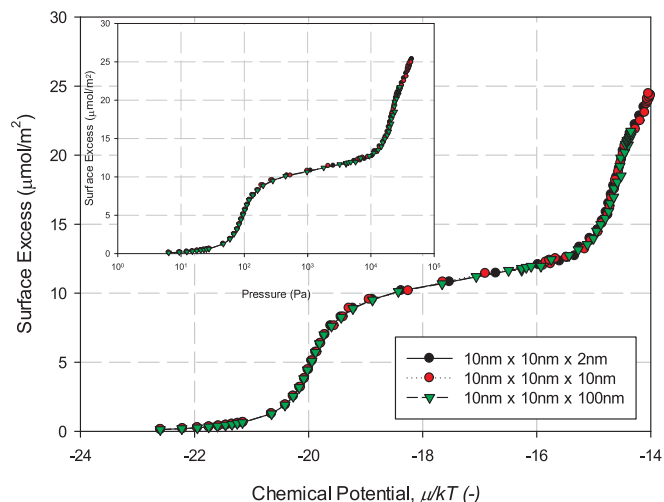


Fig. 6. Adsorption isotherm for argon on surface at 87 K for three different box sizes.

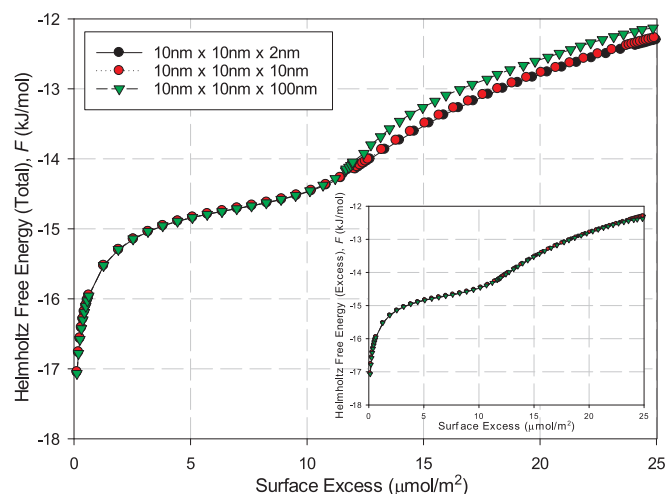


Fig. 7. Total and excess (insert) Helmholtz free energy of argon on surface at 87 K for different box sizes. The inset shows the excess Helmholtz free energy.

system becomes denser.

3.1.2. Surface tension

The surface tension in a system with two phases separated by a plane interface is most often obtained using the mechanical route of Irving, Kirkwood and Buff [19]. Here we offer an alternative route using the Helmholtz free energy of the system. In our simulation of the bulk phase of argon at 87 K, the vapour is in equilibrium with the liquid

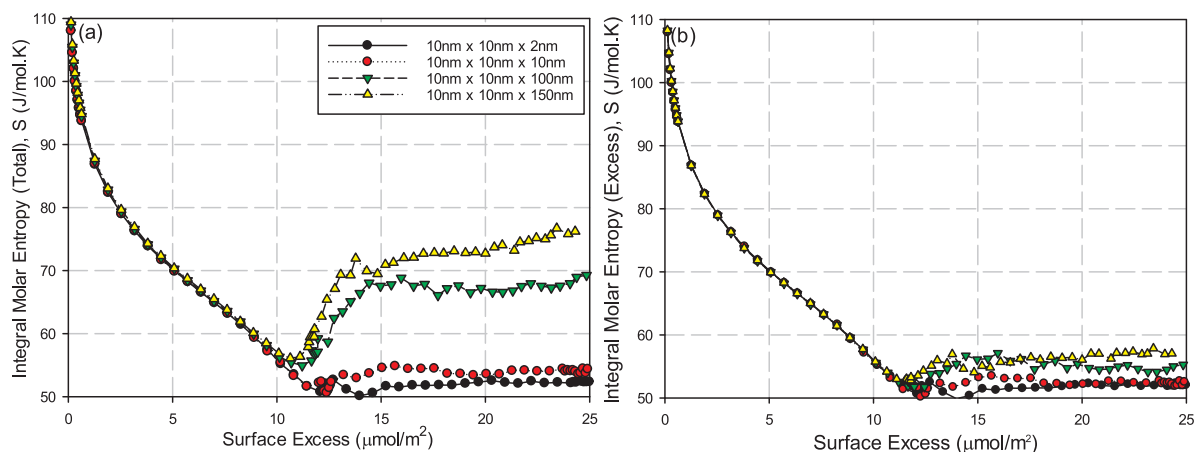


Fig. 8. (a) Total and (b) excess molar entropy of argon on surface at 87 K for different box height in the Z-direction.

phase when the chemical potential is constant (Fig. 5b). Knowing the Helmholtz free energy and using the smoothed density distribution (Appendix C) across the two phases, we determined the position of the Gibbs dividing plane at the interface, and hence the number of molecules in each phase. With the knowledge of the molar Helmholtz free energy of the pure gas phase and that of the pure liquid phase, we determined the excess Helmholtz free energy and calculated the surface tension from Eq. (18). The results are shown in Table 2, and are in good agreement with previous simulations by Goujon et al. [20] obtained using the mechanical route of Kirkwood and Buff [19]. Both values of surface tension are about 20% higher than the reported experimental data, and this is attributed by Goujon et al. [20], to the neglect of the three-body interactions.

3.2. Adsorption

We take adsorption of argon on a graphite surface as an example of the application of our method to determine the Helmholtz free energy of non-uniform adsorption systems. Fig. 6 shows plots of the excess

density versus the chemical potential for three different box sizes. These isotherms very nearly overlap, indicating that the excess amount is independent of the box size.

The integral molar Helmholtz free energies for the above three box sizes are shown in Fig. 7. With the exception of the sub-monolayer region, the molar Helmholtz free energy is smaller in larger boxes because there is a larger contribution from the molecules in the gas phase in the larger boxes. Therefore, it is informative to determine the free energy intrinsic to the adsorbed phase, by calculating the excess Helmholtz free energy using Eq. (19). The inset in Fig. 7 shows that the excess Helmholtz free energy is the same for all three box sizes.

Fig. 8a shows that the same conclusion can be reached for the molar entropy. In the sub-monolayer region, the integral molar entropy decreases to a minimum when the first adsorbate layer is complete because there is a decrease in the number of available configurations. The approach to the entropy minimum is different for different box sizes; again because there is a larger contribution from molecules in the gas phase in the larger boxes but, as found for the Helmholtz energy, the excess molar entropy is the same for all box sizes as shown in Fig. 8b.

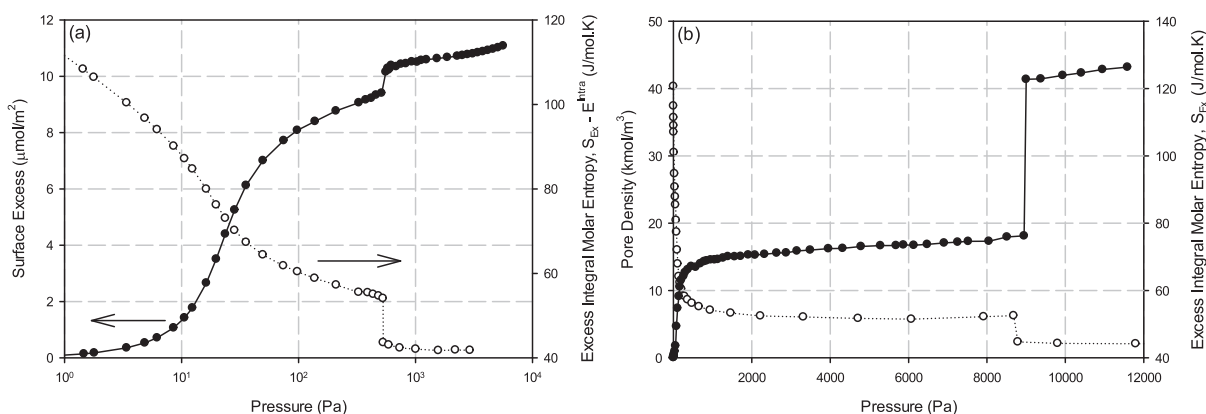


Fig. 9. Adsorption isotherm (filled) and integral molar entropy (unfilled) of (a) nitrogen on surface at 77 K and (b) argon in 2 nm open ended slit pore at 87 K.

The minimum molar entropy at the monolayer density is 53 J/mol/K, which is slightly lower than the molar entropy of the pure liquid (54.7 J/mol/K), suggesting that the adsorbed phase behaves like a compressed liquid at completion of the first layer. Beyond the monolayer region, the molar entropy increases and becomes asymptotic to the molar entropy of the pure liquid, indicating that the multi-layered adsorbed film resembles the bulk liquid.

Nitrogen adsorbed on graphite at 77 K has a first order transition from a supercritical state to a commensurate solid in the first layer, which is also exhibited for adsorption in a slit shaped mesopore, prior to a first order capillary condensation. A method for the determination of the Helmholtz free energy, when the chemical potential plot vs the number of molecules shows a first order transition, is given in Appendix D. The isotherm for nitrogen adsorption on a graphite surface is presented in Fig. 9a, and shows a first order transition when the surface density reaches $9.3 \mu\text{mol}/\text{m}^2$. At this density, the excess molar entropy is 43 J/mol/K, compared to the molar entropy of the bulk liquid nitrogen of 47 J/mol/K, indicating that the first layer is in a supercritical state prior to the transition. The first order transition changes the state of the adsorbate from a supercritical state to a commensurate state, as evidenced by the density of $10.54 \mu\text{mol}/\text{m}^2$, theoretically calculated for a commensurate packing on graphite [35]. This transition is associated with a sharp decrease in the entropy, to 31 J/mol/K, a value that is much lower than that of the bulk liquid. In fact, a sharp decrease in the entropy occurs whenever there is a first order transition in the adsorbate. To illustrate this point, we present the adsorption isotherm for argon in a 2 nm open-ended graphitic slit pore in Fig. 9b, which shows that a capillary condensation (first order transition) occurs at 8.5 kPa. This transition has been discussed in a previous paper [30]. The

interesting feature here is that the molar excess entropy decreases sharply to 42 J/mol/K, which is well below the molar entropy of the bulk liquid. This indicates that the argon is strongly adsorbed in a 2 nm pore and forms a highly compressed liquid which is corroborated by the fact that the pore density is much higher than the bulk liquid density ($35 \text{ kmol}/\text{m}^3$).

4. Conclusion

In this paper we have presented an efficient scheme for the accurate determination of the chemical potential, the molar Helmholtz free energy and the molar entropy using a kinetic Monte Carlo method. This scheme is discussed in detail for argon and further illustrated in the Appendix for a range of molecules. The molar Helmholtz free energy and the molar entropy for the bulk fluids agree well with results from the relevant equations of state. In addition, using the excess Helmholtz free energy at the Gibbs interface, we are able to determine the surface tension when two phases co-exist, offering an alternative to the mechanical route. The concept of excess used to determine the amount adsorbed in adsorption systems is applied to determine the excess (adsorbed) molecular Helmholtz free energy and entropy, which are found to be independent of the size of the simulation box for a given pressure. Finally, we explored the application of the excess entropy to adsorption systems in a mesopore where there is a phase transition.

Acknowledgement

This work is supported by the Australian Research Council (DP160103540).

Appendix A.

The fundamental equation of thermodynamics is:

$$dE = TdS - PdV + \mu dN \quad (\text{A1.1})$$

The Helmholtz free energy and its derivative are

$$A = E - TS \quad (\text{A1.2})$$

$$dA = dE - TdS - SdT \quad (\text{A1.3})$$

Substitute Eq. (A1.1) into Eq. (A1.3):

$$dA = TdS - PdV + \mu dN - TdS - SdT \quad (\text{A1.4.1})$$

$$dA = -SdT - PdV + \mu dN \quad (\text{A1.4.2})$$

Thus, at constant T , V and N (canonical ensemble), we have $dA = 0$. Therefore, the minimum Helmholtz free energy is attained at equilibrium in a canonical ensemble. In addition, we take the advantage of the accurate chemical potential determined from canonical *kMC*:

$$\left(\frac{dA}{dN} \right)_{V,T} = \mu \quad (\text{A1.5})$$

Therefore, we can determine the Helmholtz free energy by integration Eq. (A1.5) with respect to N

$$\int_{A_1}^A dA = \int_1^N \mu(N, V, T) dN \quad (\text{A1.6})$$

where A_1 is the Helmholtz free energy of one molecule, given by

$$A_1 = kT [\ln(\Lambda^3/V) - 1] \quad (\text{A1.7})$$

Appendix B.

Here, we show the chemical potential, the Helmholtz free energy and the integral molar entropy of sub-critical for supercritical of nitrogen (Fig. 10), carbon dioxide (Fig. 11) and ammonia (Fig. 12). These figures show that the simulation results (with long range correction added to the molecular energy) agree well with their respective EOS. Note that the entropy presented here for the polyatomic system, excludes the kinetic energy (Eq. (15)).

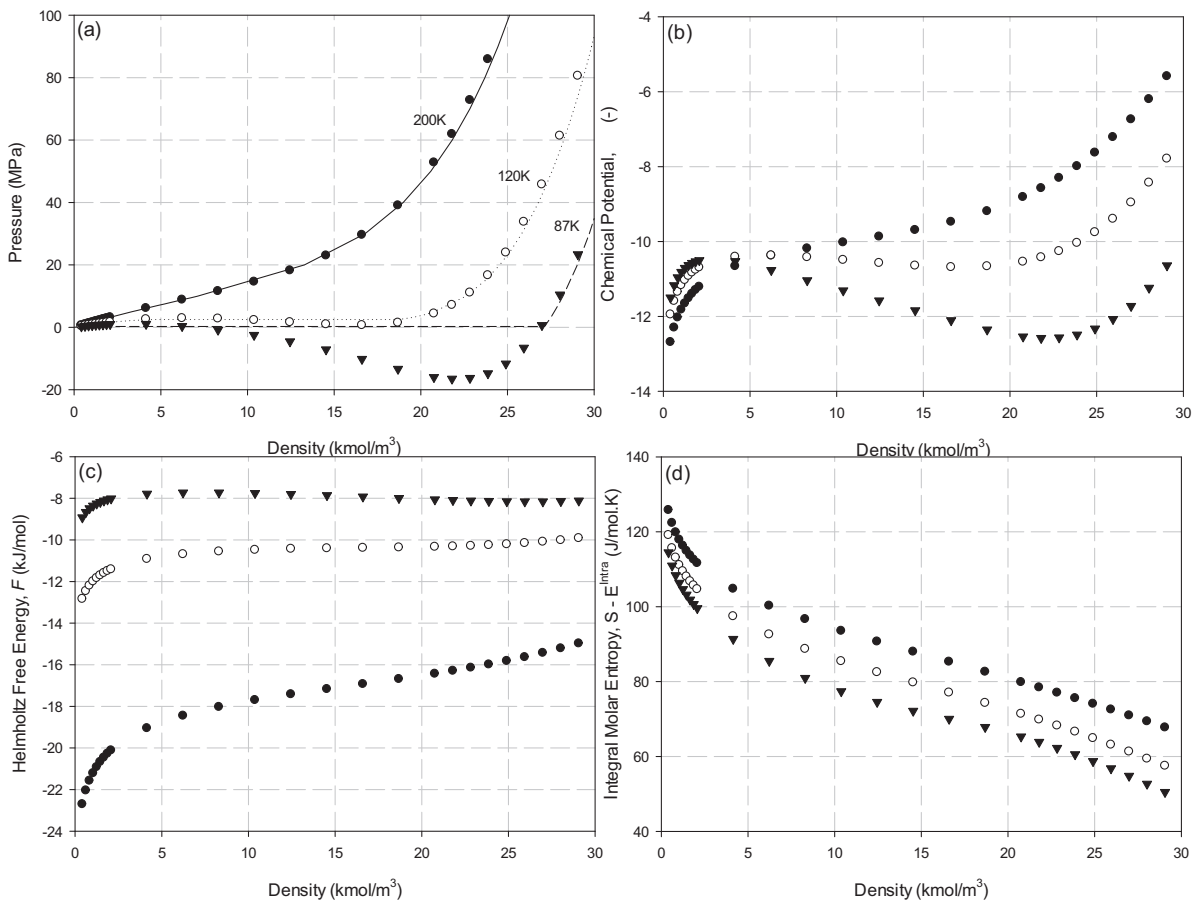


Fig. 10. (a) Pressure, (b) chemical potential, (c) Helmholtz free energy and (d) integral molar entropy of Nitrogen at 87 K, 120 K and 200 K. The lines represent EOS of Span et al. [36] and the symbols represents the simulated data.

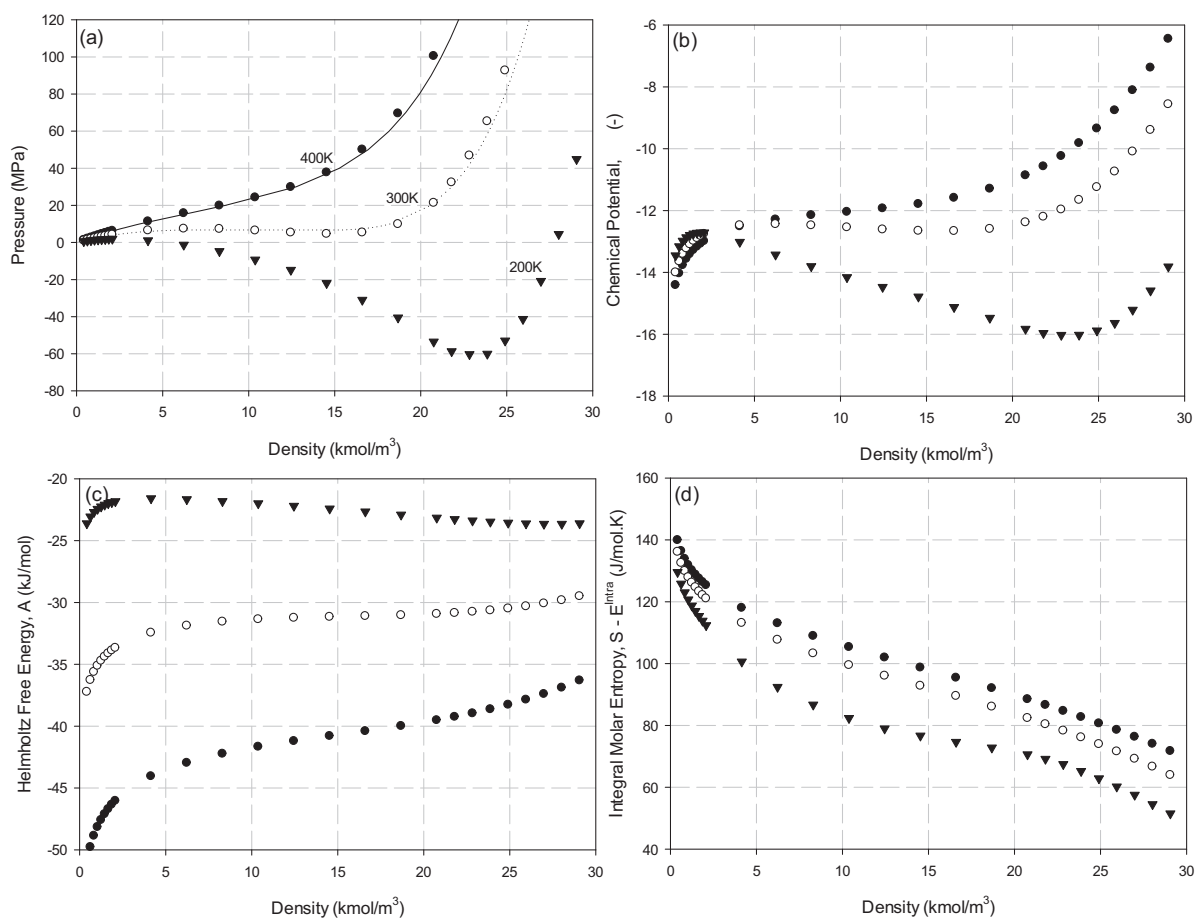


Fig. 11. (a) Pressure, (b) chemical potential, (c) Helmholtz free energy and (d) integral molar entropy of Carbon dioxide at 200 K, 300 K and 400 K. The lines represent EOS of Span and Wagner [35] and the symbols represent the simulated data.

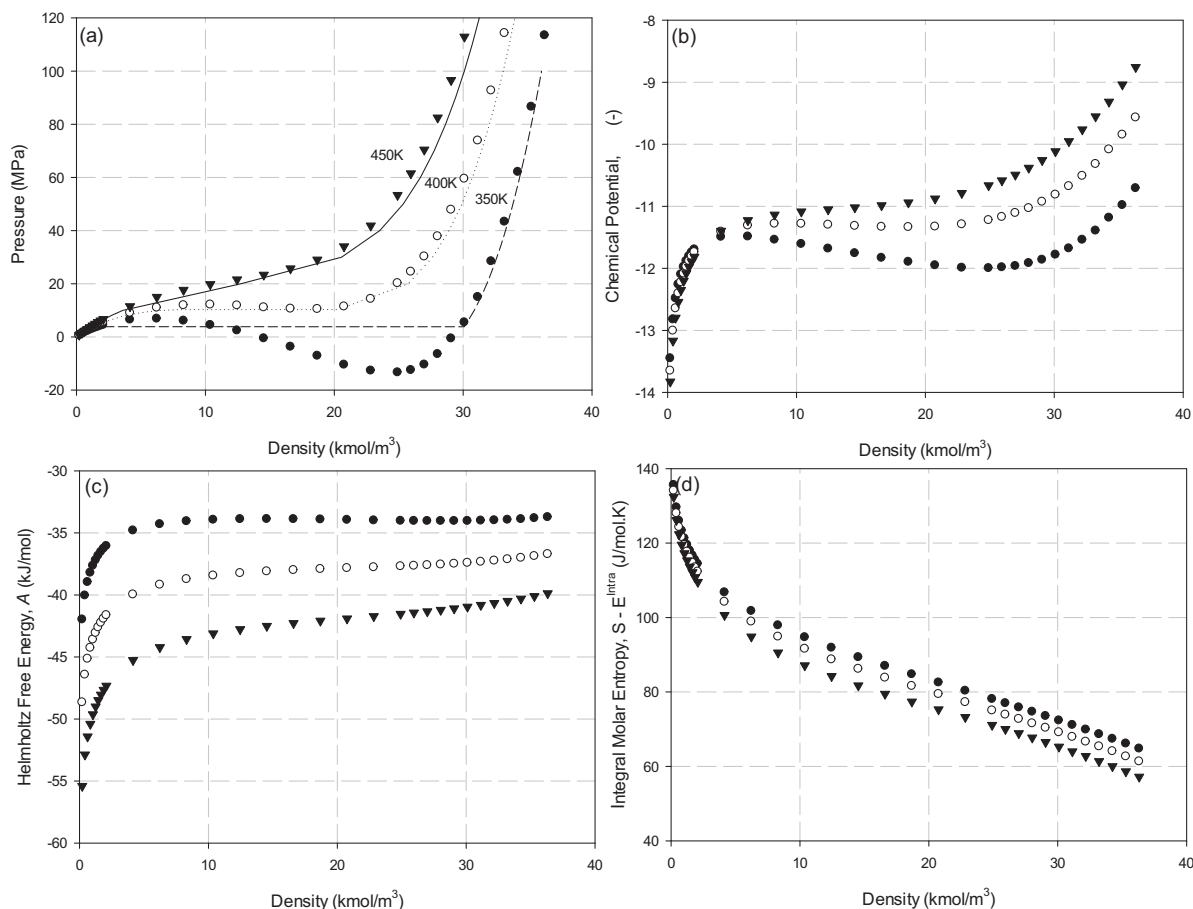


Fig. 12. (a) Pressure, (b) chemical potential, (c) Helmholtz free energy and (d) integral molar entropy of Ammonia at 350 K, 400 K and 450 K. The lines represent EOS of Roth et al. [37] and the symbols represent the simulated data.

Appendix C.

The local density distribution (LDD) for three selected points in Fig. 5d, are presented in Fig. 13. With the aid of the LDD, the number of molecules in the vapour and liquid phase can be determined by applying the concept of equal areas, as implemented in the determination of the Gibbs dividing surface. The number of molecules in those phases is then applied in Eq. (17b) to calculate the surface tension.

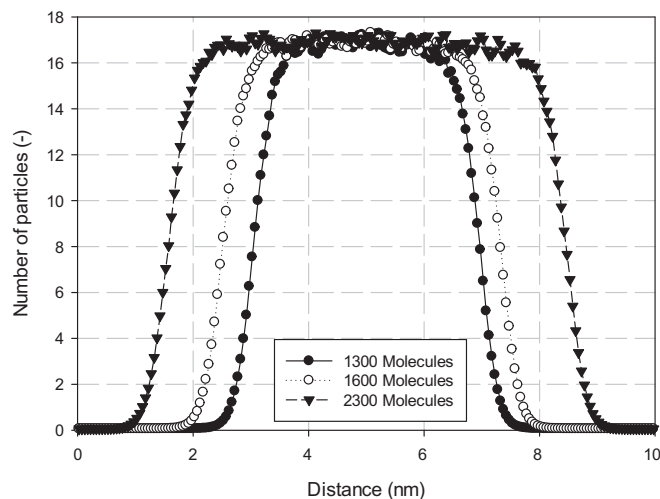


Fig. 13. Local density distribution for vapour-liquid equilibrium of argon at 87 K.

Appendix D.

The Helmholtz free energy is calculated formally by integrating the chemical potential with respect to the number of molecule in the region where the function is continuous, i.e.

$$A = \int_1^N \mu dN \quad (\text{A4.1})$$

Across the first order transition, the change in the Helmholtz free energy is simply:

$$\Delta A = \mu_{eq}(N_{After} - N_{Before}) \quad (\text{A4.2})$$

where μ_{eq} is the chemical potential at the first order transition, and N_{After} and N_{Before} are the numbers of molecules after and before the transition. Therefore, the free energy is simply the sum of Eqs. (A4.1) and (A4.2).

References

- [1] D. Frenkel, B. Smit, Understanding molecular simulation: from algorithms to applications, second ed., Academic Press, San Diego, 2002, p. 638.
- [2] E.A. Ustinov, D.D. Do, Application of kinetic Monte Carlo method to equilibrium systems: vapour-liquid equilibria, *J. Colloid Interface Sci.* 366 (1) (2012) 216–223.
- [3] E.A. Ustinov, D.D. Do, Two-dimensional order-disorder transition of argon monolayer adsorbed on graphitized carbon black: kinetic Monte Carlo method, *J. Chem. Phys.* (2012) 136 134702/1-134702/10.
- [4] E.A. Ustinov, D.D. Do, Thermodynamic analysis of ordered and disordered monolayer of argon adsorption on graphite, *Langmuir* 28 (25) (2012) 9543–9553.
- [5] D.T. Gillespie, Exact stochastic simulation of coupled chemical reactions, *J. Phys. Chem.* 81 (25) (1977) 2340–2361.
- [6] C.C. Battaile, The kinetic Monte Carlo method: foundation, implementation, and application, *Comput. Methods Appl. Mech. Eng.* 197 (41–42) (2008) 3386–3398.
- [7] E.A. Ustinov, D.D. Do, Two-dimensional order-disorder transition of argon monolayer adsorbed on graphitized carbon black: kinetic Monte Carlo method, *J. Chem. Phys.* 136 (13) (2012) 134702.
- [8] V.T. Nguyen, D.D. Do, D. Nicholson, E.A. Ustinov, Application of the kinetic Monte Carlo method in the microscopic description of argon adsorption on graphite, *Mol. Phys.* 110 (18) (2012) 2281–2294.
- [9] V.T. Nguyen, S. Tan, D.D. Do, D. Nicholson, Application of kinetic Monte Carlo method to the vapour-liquid equilibria of associating fluids and their mixtures, *Mol. Simul.* (2015) 1–13.
- [10] C. Fan, D.D. Do, D. Nicholson, E.A. Ustinov, Chemical potential, Helmholtz free energy and entropy of argon with kinetic Monte Carlo simulation, *Mol. Phys.* 112 (1) (2013) 60–73.
- [11] Widom, Some topics in the theory of fluids, *J. Chem. Phys.* 39 (11) (1963) 2808–2812.
- [12] B. Widom, Potential-distribution theory and the statistical mechanics of fluids, *J. Phys. Chem.* 86 (6) (1982) 869–872.
- [13] E.A. Ustinov, D.D. Do, Application of kinetic Monte Carlo method to equilibrium systems: vapour-liquid equilibria, *J. Colloid Interface Sci.* 366 (1) (2012) 216–223.
- [14] J. Puiasset, Grand potential, helmholtz free energy, and entropy calculation in heterogeneous cylindrical pores by the grand canonical Monte Carlo simulation method, *J. Phys. Chem. B* 109 (1) (2005) 480–487.
- [15] S. Prestipino, P. Giaquinta, Statistical entropy of a lattice-gas model: multiparticle correlation expansion, *J. Statist. Phys.* 96 (1–2) (1999) 135–167.
- [16] S.D. Hong, Evaluation of the excess free energy for two-center-Lennard-Jones liquids using the bent effective acceptance ratio, *Bull. Korean Chem. Soc.* 21 (7) (2000) 697–700.
- [17] H. Do, J.D. Hirst, R.J. Wheatley, Rapid calculation of partition functions and free energies of fluids, *J. Chem. Phys.* 135 (17) (2011) 174105–174107.
- [18] J.P.R.B. Walton, D.J. Tildesley, J.S. Rowlinson, J.R. Henderson, The pressure tensor at the planar surface of a liquid, *Mol. Phys.* 48 (6) (1983) 1357–1368.
- [19] J.G. Kirkwood, F.P. Buff, The statistical mechanical theory of surface tension, *J. Chem. Phys.* 17 (1949) 338–343.
- [20] F. Goujon, P. Malfreyt, D.J. Tildesley, The gas-liquid surface tension of argon: a reconciliation between experiment and simulation, *J. Chem. Phys.* 140 (24) (2014) 244710.
- [21] D. Nicholson, N.G. Parsonage, Computer Simulation and the Statistical Mechanics of Adsorption, Academic Press, London, 1982.
- [22] L. Herrera, D.D. Do, D. Nicholson, A Monte Carlo integration method to determine accessible volume, accessible surface area and its fractal dimension, *J. Colloid Interface Sci.* 348 (2) (2010) 529–536.
- [23] A. Michels, H. Wijk, H. Wijk, Isotherms of argon between 0°C and 150°C and pressures up to 2900 atmospheres, *Physica* 15 (7) (1949) 627–633.
- [24] J.J. Potoff, J.I. Siepmann, Vapor – liquid equilibria of mixtures containing alkanes, carbon dioxide and nitrogen, *AIChE* 47 (7) (2001) 1676–1683.
- [25] T. Kristof, J. Vorholz, J. Liszi, B. Rumpf, G. Maurer, A simple effective pair potential for the molecular simulation of the thermodynamic properties of ammonia, *Mol. Phys.* 97 (10) (1999) 1129–1137.
- [26] W.A. Steele, The physical interaction of gases with crystalline solids: I. Gas-solid energies and properties of isolated adsorbed atoms, *Surf. Sci.* 36 (1) (1973) 317–352.
- [27] W.E. Carlos, M.W. Cole, Interaction between a He atom and a graphite surface, *Surf. Sci.* 91 (1) (1980) 339–357.
- [28] L. Prasetyo, S. Tan, Y. Zeng, D.D. Do, D. Nicholson, An improved model for N₂ adsorption on graphitic adsorbents and graphitized thermal carbon black—The importance of the anisotropy of graphene, *J. Chem. Phys.* 146 (18) (2017) 184702.
- [29] M.J. Bojan, W.A. Steele, Computer simulation of physisorption on a heterogeneous surface, *Surf. Sci.* 199 (3) (1988) L395–L402.
- [30] J.K. Johnson, J.A. Zollweg, K.E. Gubbins, The Lennard-Jones equation of state revisited, *Mol. Phys.* 78 (3) (1993) 591–618.
- [31] J.-P. Hansen, L. Verlet, Phase transitions of the Lennard-Jones system, *Phys. Rev.* 184 (1969) 151.
- [32] M. Schrader, P. Virnau, K. Binder, Simulation of vapor-liquid coexistence in finite volumes: a method to compute the surface free energy of droplets, *Phys. Rev. E* 79 (6) (2009) 061104.
- [33] S. Tan, L. Prasetyo, Y. Zeng, D.D. Do, D. Nicholson, On the consistency of NVT, NPT, μ VT and Gibbs ensembles in the framework of kinetic Monte Carlo – Fluid phase equilibria and adsorption of pure component systems, *Chem. Eng. J.* 316 (2017) 243–254.
- [34] E.W. Lemmon, S.G. Penoncello, The surface tension of air and air component mixtures, Springer US, Boston, MA, 1994, pp. 1927–1934.
- [35] R. Span, W. Wagner, A new equation of state for carbon dioxide covering the fluid region from the triple-point temperature to 1100 K at pressures up to 800 MPa, *J. Phys. Chem. Ref. Data* 25 (6) (1996) 1509–1596.
- [36] R. Span, E.W. Lemmon, R.T. Jacobsen, W. Wagner, A. Yokozeki, A reference equation of state for the thermodynamic properties of nitrogen for temperatures from 63.151 to 1000 K and pressures to 2200 MPa, *J. Phys. Chem. Ref. Data* 29 (6) (2000) 1361–1433.
- [37] R. Tillner-Roth, F. Harms-Watzenberg, H. Baehr, Eine neue fundamentalgleichung für ammoniak, *DKV Tagungsbericht* 20 (1993) 167–180.